

Younas Khan — Research Fellow Candidate – Edge AI Security

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Professional Summary

PhD candidate in Machine Learning with **7+ years of research and teaching experience** in AI, privacy, and security of image data. Expertise in adversarial machine learning, privacy-preserving image anonymization, and federated learning. Proficient in Python-based deep learning (PyTorch, TensorFlow) with applications to **Edge AI, Trustworthy AI, and security-driven model evaluation**. Published researcher with **100+ citations**, currently developing robust AI techniques against adversarial and hardware-constrained attacks.

Education

Universitat Rovira i Virgili

PhD in Machine Learning

Tarragona, Spain

2022–2025

MSCA Fellowship • Thesis: "AI-Based Utility-Preserving Face Image Anonymization" • Defense: October 2025

National University of Sciences & Technology

Islamabad, Pakistan

M.Sc. Computer Software Engineering

2015–2019

Thesis: "Deep Learning based Decision Support Systems" • Focus: Deep Learning, Data Engineering

Institute of Management Sciences

Peshawar, Pakistan

B.Sc. Computer Science

2010–2015

Research Experience

Universitat Rovira i Virgili

Tarragona, Spain

MSCA Predoctoral Fellow

2022–Present

- Developed adversarial attacks (iterative similarity momentum attack) for image privacy and robustness evaluation.
- Built deep learning pipelines for image anonymization and privacy-preserving transformations in latent space.
- Published in AI privacy, federated learning, and security-related venues.
- Collaborated with European research teams on privacy-aware and trustworthy AI.

Capital University of Science & Technology

Islamabad, Pakistan

Lecturer & Research Supervisor

Feb 2020–Jun 2022

- Taught Machine Learning, Cryptography, Programming, and HCI.
- Supervised 15+ undergraduate projects in deep learning, adversarial machine learning, and data security.
- Conducted applied research in emerging AI technologies.

Technical Skills

Programming: Python (NumPy, Pandas, PyTorch, TensorFlow, Scikit-learn), C++, SQL

AI Security: Adversarial Machine Learning, Privacy-Preserving AI, Trustworthy AI, Federated Learning

Edge/Embedded AI: Model Compression, Resource-Constrained ML, Data-efficient Processing

Data Processing: Large-scale Dataset Curation, Image Processing, Statistical Validation

Research Tools: Git, Jupyter, Academic Publishing

Key Research Projects

PhD Research

Robust Face Image Anonymization

2024–2025

Designed adversarially robust anonymization methods using latent-space manipulation.

Achieved privacy preservation under PSNR/perceptual quality constraints while resisting re-identification attacks.

PhD Research

Adversarial Robustness in Edge ML

2024

Proposed iterative adversarial attack pipeline to evaluate robustness of ML models in resource-constrained conditions.

Integrated similarity-based loss and PSNR constraints for realistic perturbations.

M.Sc. Thesis

Coronary Artery Disease Detection System

2018–2019

Developed ML models for cardiovascular disease diagnosis using UCI dataset.
Applied optimization and feature selection to enhance predictive accuracy.

Research Project

Federated Learning Literature Analysis

2023–2024

Systematically analyzed 200+ research papers in federated learning.
Identified security, privacy, and communication challenges in distributed ML.

Selected Publications

Submitted: Younas Khan, A. Monreale, C. Metta, D. Sánchez, J. Domingo-Ferrer. "Explainability-Driven Image Anonymization in Latent Space." *Knowledge Discovery and Data Mining*.

Submitted: Younas Khan, D. Sánchez, J. Domingo-Ferrer. "Perceptually Robust Iterative Similarity Momentum Attack." *Engineering Applications of Artificial Intelligence*.

2025: R. Haffar, D. Sánchez, **Younas Khan**, J. Domingo-Ferrer. "Explainability-Driven Incremental Image Anonymization." *To Appear in: Transactions on Data Privacy*.

2024: Younas Khan, D. Sánchez, J. Domingo-Ferrer. "Federated Learning-based Natural Language Processing: A Systematic Literature Review." *Published in: Artificial Intelligence Review*.

Core Competencies

Research Skills: Adversarial Robustness, Secure Edge AI, Privacy-preserving ML, Federated Learning

Communication: Technical Writing, Research Presentation, International Collaboration

Languages: English (Fluent) • Spanish (Intermediate) • Urdu (Native)

Additional Information

Focus: AI Security, Edge AI, and Trustworthy ML under constrained hardware settings

Location: Based in Tarragona, Spain (close to Barcelona) • Available for relocation to Barcelona

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August 25, 2025

Hiring Committee
Queen's University Belfast
University Road, Belfast, BT7 1NN
United Kingdom

Dear Hiring Committee,

Application for Research Fellow in Edge AI Security (Job Reference: 25/112741)

Securing machine learning systems in resource-constrained environments is at the heart of my research. Over the past seven years, I have developed expertise in adversarial machine learning, privacy-preserving AI, and robustness evaluation—skills that align directly with the challenges of edge AI security. My PhD research at Universitat Rovira i Virgili, funded through an MSCA Fellowship, focuses on adversarial attacks, defense strategies, and latent-space transformations to preserve both utility and privacy under constrained hardware conditions. This work translates directly to the design and evaluation of secure AI systems on embedded and edge platforms.

In my doctoral work, I proposed iterative adversarial pipelines that simulate realistic attacks under performance and perceptual constraints, providing frameworks for evaluating robustness in edge and federated settings. I have also published a systematic review in *Artificial Intelligence Review* analyzing 200+ papers in federated learning, where I identified open challenges in privacy, security, and distributed deployment. Beyond my PhD, my teaching and supervision experience includes machine learning, cryptography, and adversarial robustness projects—demonstrating my ability to combine theoretical insight with applied system-level design.

What I bring to Queen's University Belfast:

- **Research expertise:** Adversarial ML, privacy-preserving AI, and robustness analysis aligned with secure embedded AI systems.
- **Technical proficiency:** Python (PyTorch, TensorFlow, scikit-learn), large-scale ML pipelines, and statistical foundations in optimization and evaluation.
- **Security perspective:** Experience in evaluating attack surfaces and developing countermeasures under resource and perceptual constraints.
- **Collaborative track record:** Publications in leading AI journals, international project collaborations across Europe, and supervision of ML security-focused student projects.

I am particularly motivated by this position at Queen's University Belfast because it sits at the intersection of my core expertise: adversarial robustness, privacy-aware AI, and the deployment of secure machine learning in edge environments. I am eager to contribute to the development of secure AI accelerators, performance-security trade-off frameworks, and methods to safeguard AIoT deployment.

Thank you for considering my application. I look forward to the opportunity to discuss how my research background can contribute to advancing secure Edge AI at Queen's University Belfast.

Sincerely,

Younas Khan